

Section 26: EtherChannel

Layer: L2 | Reference: Odom Vol1 Ch10: RSTP and EtherChannel Concepts | *Original study notes — CCNA 200-301*

EtherChannel bundles multiple physical switch links into one logical link, increasing bandwidth and providing redundancy without STP blocking any of the member ports (since STP sees the bundle as a single logical link).

Key Concepts

- EtherChannel combines 2-8 physical ports into a single logical Port-Channel interface
- LACP (Link Aggregation Control Protocol, IEEE 802.3ad / 802.1AX) is the open-standard negotiation protocol
- PAgP (Port Aggregation Protocol) is the older Cisco-proprietary equivalent
- Static EtherChannel ('mode on') bundles links with no negotiation protocol at all - both sides must be manually set identically, with no verification
- LACP modes: Active (sends LACP packets to initiate negotiation) and Passive (waits for the other side to initiate)
- Valid LACP combinations: Active+Active, or Active+Passive. Passive+Passive will NOT form a channel since neither side initiates
- PAgP modes: Desirable (initiates) and Auto (waits) - same logic as LACP active/passive; Auto+Auto will not form a channel
- All member ports must match: speed, duplex, trunk/access mode, allowed VLANs, and native VLAN - mismatches prevent the channel from forming or cause it to fail
- Load balancing across member links is configurable (e.g., by source/destination MAC, IP, or port) but is NOT per-packet round robin - a single flow stays on one physical link to preserve ordering
- EtherChannel is configured first at the physical interface level (channel-group), then the resulting Port-Channel interface is configured for trunking/VLANs

Command Reference

Command	Purpose
<code>channel-group <n> mode {active passive desirable auto on}</code>	Add a physical interface to an EtherChannel bundle with the chosen negotiation mode
<code>interface port-channel <n></code>	Enter configuration mode for the logical bundled interface
<code>port-channel load-balance <method></code>	Set the algorithm used to distribute traffic across member links
<code>show etherchannel summary</code>	Quick view of channel groups, member ports, and their status (flags like SU = bundled)
<code>show interfaces port-channel <n></code>	Detailed status of the logical Port-Channel interface

Configuration Walkthrough

Create an LACP EtherChannel between two switches (active/active)

```
SW1(config)#interface range GigabitEthernet0/1-2
SW1(config-if-range)#channel-group 1 mode active
SW1(config)#interface port-channel 1
SW1(config-if)#switchport mode trunk
```

Matching configuration on the far-end switch

```
SW2(config)#interface range GigabitEthernet0/1-2
SW2(config-if-range)#channel-group 1 mode active
SW2(config)#interface port-channel 1
SW2(config-if)#switchport mode trunk
```

Worked Scenario

Q: Two switches are cabled with two redundant links and channel-group commands are configured on both ends, but the EtherChannel never comes up. SW1 uses 'mode active' and SW2 uses 'mode auto'. What's wrong?

Mode mismatch across protocols. 'Active' is an LACP mode; 'auto' is a PAgP mode - they don't speak the same negotiation protocol to each other, so no channel forms. Both switches must use the same protocol family (both LACP: active/passive, or both PAgP: desirable/auto), and at least one side must be in the initiating mode (active or desirable) since passive+passive (or auto+auto) will also fail to form a channel.

Common Mistakes / Gotchas

- Mixing LACP and PAgP modes between the two ends of a bundle
- Forgetting that passive+passive (or auto+auto) never initiates negotiation
- Leaving member ports individually configured for different VLANs/trunk settings than the rest of the bundle
- Assuming EtherChannel does true per-packet load balancing - it balances per-flow, not per-packet

Exam Tips

- ★ Memorize valid LACP pairings: active-active OK, active-passive OK, passive-passive FAILS
- ★ Same logic for PAgP: desirable-desirable OK, desirable-auto OK, auto-auto FAILS
- ★ All member port settings (speed/duplex/trunk config) must match exactly before attempting to bundle